10.1 [503, 087, 512, 061, 908, 170, 897, 275, 653, 426]

(1) 直接插入排序

① [087, 503, 512, 061, 908, 170, 897, 275, 653, 426]

② [087, 503, 512, 061, 908, 170, 897, 275, 653, 426]

③ [061, 087, 503, 512, 908, 170, 897, 275, 653, 426]

④ [061, 087, 503, 512, 908, 170, 897, 275, 653, 426]

⑤ [061, 087, 170, 503, 512, 908, 897, 275, 653, 426]

⑥ [061, 087, 170, 503, 512, 897, 908, 275, 653, 426]

⑦ [061, 087, 170, 275, 503, 512, 897, 908, 653, 426]

⑧ [061, 087, 170, 275, 503, 512, 653, 897, 908, 426]

⑨ [061, 087, 170, 275, 426, 503, 512, 653, 897, 908]

(2) 分组 (503, 170), (087, 897), (512, 275), (061, 653), (908, 426)
d=5 结束状态.

[170, 087, 275, 061, 426, 503, 897, 512, 653, 908]

(3)

① [(426, 087, 275, 061, 170), 503, (897, 908, 653, 512)]

② [(170, 087, 275, 061), 426, 503, (512, 653), 897, (908)]

③ [(061, 087), 170, (275), 426, 503, 512, (653), 897, 908]

④ [061, (087), 170, 275, 426, 503, 512, 653, 897, 908]

⑤ [061, 087, 170, 275, 426, 503, 512, 653, 897, 908]

(4) 建堆 [908, 653, 897, 503, 426, 170, 512, 275, 061, 087]

① [(897, 653, 512, 503, 426, 170, 087, 275, 061), 908]

② [(653, 503, 512, 275, 426, 170, 087, 061), 897, 908]

③ [(512, 503, 170, 275, 426, 061, 087), 653, 897, 908]

④ [(503, 426, ~~087~~ 170, 275, 087, 061), 512, 653, 897, 908]

⑤ [(426, 275, 170, 061, 087), 503, 512, 653, 897, 908]

⑥ [(275, 087, 170, 061), 426, 503, 512, 653, 897, 908]

⑦ [(170, 087, 061), 275, 426, 503, 512, 653, 897, 908]

⑧ [(087, 061), 170, 275, 426, 503, 512, 653, 897, 908]

⑨ [061, 087, 170, 275, 426, 503, 512, 653, 897, 908]

(5) 归并.

① [(087, 503), (061, 512), (170, 908), (275, 897), (426, 653)]

② [(061, 087, 503, 512), (170, 275, 897, 908), (426, 653)]

③ [(061, 087, 170, 275, 503, 512, 897, 908), (426, 653)]

④ [(061, 087, 170, 275, 426, 503, 512, 653, 897, 908)]

(6)

① [170, 061, 512, 503, 653, 275, 426, 087, 897, 908]

② [503, 908, 512, 426, 653, 061, 170, 275, 087, 897]

③ [061, 087, 170, 275, 426, 503, 512, 653, 897, 908]

10.3 稳定的: (1) (5) (6)

不稳定的: (2) (3) (4)

反例. (2): [2, 2*, 1], 当 $d=2$.

排序后: [1, 2*, 2].

(3) [2, 2*, 1], 以第一个2为基准

排序后: [1, 2*, 2]

(4): [2, 2*, 1]

排序后 [1, 2*, 2]

10.6. (1) 在连续的一趟奇数项比较和一趟偶数项比较中, 没有发生任何交换动作.

(2) 单轮比较次数 $= \lfloor \frac{n}{2} \rfloor + \lfloor \frac{n-1}{2} \rfloor = n-1$

正序时: 进行一轮结束. 比较次数为 $n-1$ 次

逆序时: 需进行 $n$ 趟

$n$ 为奇数时:

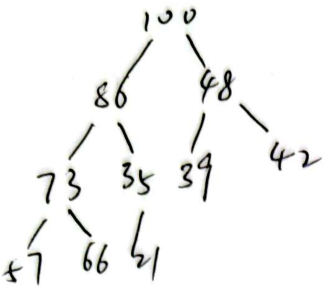
$$\underbrace{\frac{(n+1)}{2} \cdot \frac{(n-1)}{2}}_{\text{奇数轮}} + \underbrace{\frac{n-1}{2} \cdot \frac{n-1}{2}}_{\text{偶数轮}} = \frac{n^2-n}{2} = \frac{n(n-1)}{2}.$$

$n$ 为偶数时:

$$\underbrace{\frac{n}{2} \cdot \frac{n}{2}}_{\text{奇}} + \underbrace{\frac{n}{2} \cdot \frac{n-2}{2}}_{\text{偶}} = \frac{n^2-n}{2} = \frac{n(n-1)}{2}$$

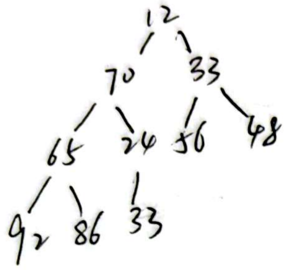
10.12

(1) [100, 86, 48, 73, 35, 39, 42, 57, 66, 21]



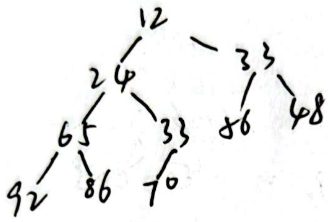
是大根堆

(2) [12, 70, 33, 65, 24, 56, 48, 92, 86, 33]

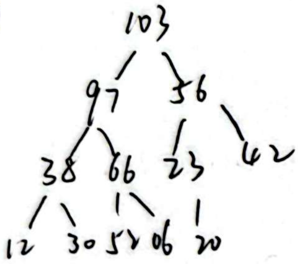


不是堆.

调整为小根堆，交换2次

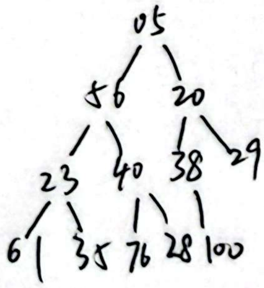


(3)



是大根堆

(4)



不是堆

调整为小根堆，交换3次